ET AI HACKATHON 2.0 | PROBLEM STATEMENT 2

SETU

Strategic Energy Trade Uncertainty

AI-powered energy supply-chain resilience for import-dependent economies.

SETU converts a source-grounded geopolitical disruption into an auditable operational decision trail: evidence, corridor risk, forecast, cascade simulation, and explainable mitigation options for human approval.

PROBLEM	WHY IT MATTERS
Import-dependent crude supply is exposed to disruption in a small number of maritime corridors.	A decision team needs a defensible response path - not a stream of alerts - when a corridor is threatened.

Working prototype | React + FastAPI + SQLite | Source evidence + simulation + human-in-the-loop decisions

1. Problem and opportunity

Turning reactive crisis response into anticipatory supply-chain resilience

India's crude-oil supply is structurally sensitive to geopolitical and maritime disruption. A disruption at a critical corridor can cascade through port access, refinery throughput, demand fulfillment, and procurement decisions. Existing monitoring tools surface information; SETU is designed to convert that information into a transparent decision sequence.

Target user: energy-security, procurement, logistics, refinery-operations, and policy teams that must evaluate a response quickly while retaining human control.

Evaluation dimension	SETU response
Innovation	A source-to-decision workflow that joins evidence extraction, a corridor risk model, forecast telemetry, Monte Carlo cascades, and Pareto mitigation options.
Business impact	Makes the effect of a corridor disruption legible before an operator commits to rerouting, reserve use, or supplier diversification.
Technical excellence	Frozen JSON contracts, deterministic scoring/orchestration, persisted results, automated tests, and a reproducible Docker path.
Scalability	Composable API layers and schema-driven contracts allow more sources, corridors, and operational data feeds to be added.
User experience	A guided incident-response workflow and evidence-first explanations reduce cognitive load during a crisis.

2. Product workflow

A clear chain of custody from source evidence to recommended response

SETU is not presented as a black-box predictor. Every stage exposes an observable artifact that a decision-maker can inspect.

Stage	System action	What the operator sees
Evidence	Ingest a source URL or GDELT signal; extract corridor, event type, severity, evidence terms, and confidence.	Article title, publisher, source link, timestamps, confidence, and matched terms.
Risk	Update corridor-level risk score through deterministic scoring.	Affected corridor and updated risk state.
Forecast	Generate short-horizon risk telemetry from available feature data.	Forecast dates and uncertainty-aware risk values.
Cascade	Propagate the corridor disruption through the crude network using Monte Carlo simulation.	Scenario identifier and downstream impact bands.
Mitigation	Rank response options using risk/time/cost trade-offs.	Pareto-oriented options, rationale, and explicit approve/reject controls.

Guided incident response: one action runs a credible reference source through the complete sequence - source evidence -> HORMUZ risk update -> forecast -> cascade -> mitigation options. It provides a repeatable, inspectable operational workflow.

3. Technical architecture

Neuro-symbolic by design: AI-assisted extraction, deterministic decisions

The system separates interpretation from decision logic. Extraction may use rules or an LLM-backed path, while risk scoring, simulation, and recommendation ranking remain deterministic and inspectable. This is deliberate: in a high-consequence supply decision, explainability and repeatability matter as much as model sophistication.

Layer	Implementation	Purpose
Experience	React, Vite, Leaflet, Recharts	Interactive dashboard, map, evidence, forecast, cascade and recommendation views.
API	FastAPI	Typed endpoints for signals, forecasts, cascades, recommendations, health, and contracts.
Intelligence	Rule/LLM extraction + deterministic risk scoring	Convert a geopolitical text signal into a structured corridor event.
Simulation	Network graph + Monte Carlo	Represent and propagate disruption through ports, refineries, and demand nodes.
Decision	Pareto-style orchestrator + HITL	Expose mitigation trade-offs; humans approve or reject the action.
Persistence	SQLite	Keep signals, forecasts, cascades, recommendations, and audit state reproducible for the prototype.
Contracts	JSON Schema + generated types	Maintain a single source of truth between frontend and backend.

4. Data credibility and explainability

Make the evidence visible before asking a user to trust the result

• URL ingestion retains **source URL, publisher, title, publication/scrape timestamps, confidence, and matched evidence terms** in the UI.

• The dashboard labels source-grounded items and de-emphasizes weak or zero-contribution seeded rows, preventing synthetic reference data from being mistaken for live evidence.

• Recommendations show **why** an option is proposed and surface its risk reduction, time penalty, and cost trade-off before approval.

• The application supports cached samples for reproducible offline operation and optional external data sources including GDELT, OFAC, EIA, and FRED as documented in the repository.

Claim	Evidence in the product
Not just a news summary	The source triggers a corridor-specific risk update, forecast, simulation, and mitigation sequence.
Not a hard-coded dashboard	The guided incident workflow performs API actions in sequence and renders the returned evidence, forecast, cascade, and recommendations.
Not autonomous procurement	Human-in-the-loop approve/reject controls remain at the final recommendation stage.
Honest uncertainty	Forecast and cascade panels use risk/impact bands; documented limitations identify the boundaries of the MVP.

5. Prototype validation

Evidence that the submitted path runs end to end

The current prototype has been validated through frontend tests, a production build, Docker startup with a healthy API, and an in-browser run of the guided incident workflow. The verified workflow completed all five stages and rendered AP evidence, a HORMUZ risk update, forecast output, a cascade scenario, and three mitigation options.

Check	Result
Frontend automated suite	22 tests passed.
Frontend production build	Passed; Vite emitted only a bundle-size advisory.
Containerised stack	Docker build completed; backend health endpoint became available; frontend served at port 5173.
Browser verification	No console errors on the dashboard; the guided incident workflow completed all 5 stages.
Repository hygiene	The private SRS build-plan working document is ignored and excluded from the intended submission commit.

6. Product walkthrough

A focused 3-4 minute explanation of the operational workflow

Time	Show	Narration
0:00-0:25	SETU landing explanation	SETU detects geopolitical supply-chain shocks, maps affected crude corridors, forecasts risk, simulates downstream impact, and recommends mitigation.
0:25-0:55	Credibility panel	A decision begins with evidence: source, confidence, terms, and timestamps are visible.
0:55-2:10	Run Incident Response Workflow	One credible reference source drives the end-to-end sequence: evidence, risk, forecast, cascade, and mitigations.
2:10-3:00	Recommendation panel	Each option shows why it is recommended, its trade-offs, and the human approval gate.
3:00-3:30	Limitations and close	This is an auditable decision-support MVP; expanded AIS, broader validation, and live operational integrations are next.

7. Responsible scope and roadmap

What SETU proves now, and what a production deployment still needs

SETU makes explicit choices about scope rather than hiding them. It is an MVP for decision support, not a replacement for maritime intelligence, price analytics, or procurement governance.

Current MVP boundary	Production extension
English-language source text	Multilingual extraction with domain validation.
No AIS vessel tracking	Integrate vessel, port congestion, and tanker-availability feeds.
Crude throughput model	Extend to refined products, refinery compatibility, and inventory policy.
Up to 7-day forecast horizon	Calibrated longer-horizon models with ongoing backtesting.
One historical backtest crisis	Multiple event windows, baselines, and independent validation.
Single-node prototype persistence	Role-based access, audit controls, event streaming, and resilient operational infrastructure.

Next milestone: connect validated AIS, supplier, refinery, and market data feeds; calibrate the network model with domain partners; and measure detection lead time and recommendation quality against real operating scenarios.

8. How to run

Reproducible local deployment

Prerequisite: Docker Desktop (Windows/macOS) or Docker Engine with Docker Compose (Linux).

```
git clone <YOUR_GITHUB_REPOSITORY_URL>
cd SETU
docker compose up --build
```

Open **http://127.0.0.1:5173**, select **Run Incident Response Workflow**, then inspect the evidence and recommendation panels. The API health endpoint is available at **http://127.0.0.1:8000/health**.

Repository documentation includes architecture, data-source notes, known limitations, validation materials, a product walkthrough script, and a video outline.

SETU: from geopolitical signal to an explainable energy-supply decision.

Thank you for evaluating our submission.